Problem Set 6

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours.

Relevant reading: MML 5.1–5.7. The book is available online for free.

Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended).

1. Suppose that

$$\boldsymbol{A} = \begin{bmatrix} 3 & 9 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 12 & 1 \\ -3 & 2 \end{bmatrix}, \quad \boldsymbol{c} = \begin{bmatrix} 7 \\ 5 \end{bmatrix},$$

   and let

$$f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{c}^\top \boldsymbol{x}, \qquad g(\boldsymbol{x}) = \boldsymbol{B}\boldsymbol{x}.$$

   Note: The matrix $\boldsymbol{A}$ above is NOT symmetric.

   Derive:

   (a) Gradient of f
   (b) Hessian of f
   (c) Jacobian of g
   (d) Gradient of $f \circ g$ *using the chain rule*
   (e) Hessian of $f \circ g$ by constructing the Jacobian of the result above

2. Derive the gradient and Hessian of the following functions:

   (a) $f(x) = \|\boldsymbol{b} - \boldsymbol{a}x\|^2$
   (b) $f(x) = \|\boldsymbol{b} - \boldsymbol{a}x\|$ where it exists
   (c) $f(\boldsymbol{x}) = \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2$

   Here, $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$ and $\boldsymbol{A} \in \mathbb{R}^{n \times m}$.

3. MML Problems: 5.2, 5.3, 5.5, 5.7

4. **[code]** Verify at least 4 derivatives above using an automatic differentiation package of your choice (Julia: ForwardDiff[1] or DifferentiationInterface[2], Python: PyTorch[3] or JAX[4], R: torch[5] or RTMB[6]). See the repository for some examples. Please do NOT use a symbolic differentiation package, like Mathematica or SymPy.

5. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).

[1] `https://github.com/JuliaDiff/ForwardDiff.jl`
[2] `https://github.com/JuliaDiff/DifferentiationInterface.jl`
[3] `https://pytorch.org`
[4] `https://github.com/jax-ml/jax`
[5] `https://torch.mlverse.org`
[6] `https://cran.r-project.org/package=RTMB`